Yella Reddy Pallerla

Senior Embedded Software Engineer

4G LTE / VoLTE | **Embedded Linux** | **FreeRTOS** | **BLE** | **Secure OTA** | **CMMI-DEV**

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Professional Summary

Senior embedded engineer with **7.3 years** building production firmware in C/C++ across RTOS, bare-metal and Embedded Linux (Yocto). Specialises in **cellular connectivity** – 4G LTE modem integration, VoLTE/IMS call handling and PPP data paths – alongside BLE system design, Linux driver and BSP work, and secure OTA. Has taken products from schematic bring-up to field deployment across Industrial IoT, telecom infrastructure, EV charging, access-control/security and defence systems, delivered within a CMMI-DEV process with formal design reviews and requirements-driven verification.

Core Competencies

Cellular & Connectivity	4G LTE modem integration (SIMCom, AT command engine, VoLTE/IMS), PPP, dual-transport Ethernet/cellular failover, MQTT (ThingsBoard), TCP/IP, HTTP/HTTPS, SNMP
Wireless / RF	BLE GAP/GATT, NimBLE multi-role central/peripheral bridging, Arduino BLE, BLE-based OTA and pairing/reconnect design, sub-GHz RF (CC1101)
Embedded Linux	Yocto BSP customization, Device Tree, kernel configuration, character driver development, U-Boot and DDR bring-up, OTA/FOTA
RTOS & Architecture	FreeRTOS, bare-metal, POSIX threads, UNIX IPC, ioctl and user–kernel interaction, OpenAMP/RPMsg concepts, multi-core AMP
Silicon	STM32 F4/F1/G0, GD32F303, ESP32 / ESP32-S3, MSP430FR2311, TI AM335x, Qualcomm WP760, SmartFusion (Cortex-M3 + FPGA), Raspberry Pi-4, Banana Pi-M5
Industrial Protocols	Modbus RTU/TCP, CAN, RS485, RS422, RS232
Security & Provisioning	PKI-based device identity, TLS/DTLS, secure NVS-backed credential storage, checksum-verified OTA
Quality & Safety	CMMI-DEV structured process – root-cause analysis (CAR), decision analysis (DAR), formal design reviews (HLD/LLD), and requirements-driven manual V&V test execution
Languages & Tools	Embedded C, C++, Java, Python; STM32CubeIDE, Keil, IAR, Code Composer Studio, PlatformIO, VS Code, Git, GDB, JTAG/SWD, logic analyser, oscilloscope

Professional Experience

Magellanic Cloud Limited, Hyderabad

Apr 2026 – Present

Senior Firmware / Embedded Linux Engineer

NED – IoT Site Surveillance & Access Control Gateway

Clients: Punnewala, Manapuram GOLD IFL

Stack: ESP32-S3, STM32G070, SIMCom A7677S (4G LTE/VoLTE), ESP-IDF, FreeRTOS, MQTT (ThingsBoard), PPP, RS-422, TLS/PKI

- Architected the ESP32-S3 gateway firmware, integrating a 4G LTE modem over USB host (dual CDC-ACM interfaces) for AT command control and PPP data connectivity; deployed to 200 sites.
- Implemented VoLTE/IMS automatic call answering with whitelist filtering and ring-count-based accept/reject logic.
- Designed dual-transport networking with Ethernet primary and automatic cellular failover/handback under a connectivity watchdog.
- Delivered an end-to-end OTA/FOTA pipeline for the STM32 sub-device – cloud package assignment, secure download, checksum verification and bootloader flashing – achieving a 100% success rate across 200 field updates.
- Built the ThingsBoard cloud integration: gateway MQTT telemetry ingestion, an RPC control surface (door unlock, OTA triggers, configuration updates) and PKI-based device identity/secure provisioning.
- Developed a reusable dashboard widget covering voltage sensing, cellular signal/SIM diagnostics and throughput testing across multi-site, multi-device deployments.

ORCHID – BLE Smart Vault / Door Access-Control System

Stack: ESP32 (ESP-IDF/NimBLE + Arduino BLE), STM32F103 (HAL), RS485, I2C, PCF8563 RTC, EEPROM, IWDG

- Designed a dual-ESP32 BLE architecture: a bridge node acting simultaneously as GATT client to the onboard device and GATT server to the mobile app, relaying commands and notifications between them.
- Root-caused and fixed a NimBLE host state-corruption defect by relocating reconnect logic from a separate FreeRTOS

task into NimBLE host callouts, eliminating dropped mobile-app connections entirely.
- Implemented MAC-address discovery with NVS persistence and automatic scan/reconnect, giving reliable pairing across power cycles; added BLE-triggered OTA and employee-ID access management.
- Developed the STM32F103 relay/interlock controller – CRC16-verified RS485 protocol, RTC and EEPROM-backed configuration, IWDG watchdog supervision – enforcing mutually exclusive lock states across vault, main-door and BM-door relays for fail-safe operation.

MSP430 Sub-GHz Wireless Sensor Network – Architecture & Integration Layer

Status: Architecture and telemetry contract defined; STM32-side integration layer and dashboard support implemented ahead of node firmware bring-up.

- Defined the end-to-end topology for battery-powered MSP430FR2311 + CC1101 sensor nodes reporting over sub-GHz RF through a wired receiving node and UART bridge, deliberately decoupling the STM32 controller from the radio.
- Specified and reserved a documented nine-channel telemetry contract so node firmware can be developed against a fixed interface rather than one negotiated during bring-up.
- Introduced a dedicated `NotWired` sentinel for unpopulated channels instead of reporting idle, preventing a non-existent safety sensor (panic, fire/smoke) from being misread downstream as a live all-clear.

Axiom Energy Conversion Pvt. Ltd., Hyderabad

Oct 2024 – Jan 2026

Senior Embedded Software Engineer

On-Board EV Charger Firmware (1.5 kW)

Clients: Exide Industries, Pinnacle Power Systems
Stack: GD32F303CCT6, Embedded C/C++, ADC, PWM, UART

- Designed modular firmware architecture and implemented a 58.4 V / 25 A CC–CV charging algorithm as a state machine, shipped in 500 units.
- Developed ADC-based feedback and PWM control loops holding ±1% output voltage and ±2% current regulation across line and load.
- Integrated protection mechanisms for over-voltage, over-current, thermal and short-circuit conditions.
- Led hardware bring-up, debugging and production deployment.

Lineage Power Systems Pvt. Ltd., Bangalore

Dec 2020 – Sep 2024

Senior Firmware / Embedded Linux Engineer

4G-LPRTU Wireless Gateway & Power Grid Automation Controller

Clients: RJIL, BSNL, IGT
Stack: Qualcomm WP760, Yocto Linux, STM32F407, FreeRTOS, CycloneTCP, MQTT, Modbus, CAN

- Developed multi-threaded Linux user-space applications in C/C++ with socket and MQTT modules for remote telemetry across 2000 towers/sites of telecom and power-grid infrastructure.
- Performed full board bring-up: power sequencing validation, U-Boot configuration, DDR initialization and kernel boot debugging; customized Device Tree and validated Ethernet, UART, SPI, I2C and GPIO interfaces.
- Designed the OTA firmware update workflow and debugged issues across both user space and kernel space.
- Built a Linux character device driver (open/read/write/ioctl) on TI AM335x for hardware abstraction and user–kernel communication, integrated into the gateway firmware stack.
- Architected FreeRTOS-based controller firmware with multitasking and synchronization, integrating Modbus RTU and CAN stacks over a TCP/IP backhaul.

Serveen Software Solutions (DRDO–RCI), Hyderabad

Nov 2019 – Mar 2020

Software Engineer – Defense Systems

- Developed deterministic firmware for the Rustom-II defence control system on SmartFusion (Cortex-M3 + FPGA): timing-critical PWM control logic and MIL-STD-1553B protocol support under strict real-time constraints.

Hi-Q Electronics, Hyderabad

Aug 2018 – Aug 2019

Embedded Engineer

- Built microcontroller-based industrial control modules for PLC integration and factory automation – UART, SPI and I2C peripheral drivers, Modbus communication, board bring-up and system validation.

Education

B.Tech – Electronics and Communication Engineering **2013 – 2017**

Advanced Embedded Systems Training – Vector India, Hyderabad

Languages

English | Kannada | Telugu | Hindi